



EAST WEST UNIVERSITY

Assignment Report

DSA 5021: Time Series Analysis

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Forecasting Carbon Monoxide Concentrations Using Time Series and Machine Learning Forecasting Models

Abstract: Air pollution, particularly carbon monoxide (CO), poses significant health risks, making accurate forecasting essential for environmental monitoring and policy decisions. This study applied statistical and machine learning approaches to predict daily CO concentrations using a real-world dataset. After thorough preprocessing, including missing value imputation, outlier removal, and STL decomposition, three models were implemented: ARIMA, Holt-Winters exponential smoothing, and Random Forest Regressor. The models were evaluated using MAE, RMSE, and MAPE. Among the three models, the Random Forest Regressor achieved the highest accuracy, with a MAE of 0.181, RMSE of 0.244, and MAPE of 4.64%, outperforming both ARIMA and Holt-Winters in capturing linear and nonlinear dynamics of CO concentrations. The findings demonstrate that data-driven forecasting methods can provide reliable estimates for pollution monitoring, particularly where real-time sensors are limited. Overall, the study highlights the utility of combining classical time series methods with modern machine learning techniques to enhance environmental forecasting and inform public health interventions.

I. Introduction

Air pollution is one of the most pressing environmental and public health challenges worldwide. Among various pollutants, carbon monoxide (CO) is particularly harmful due to its ability to bind with hemoglobin in the blood, reducing oxygen transport and leading to severe health problems such as headaches, dizziness, and, in extreme cases, death. According to the World Health Organization (WHO), exposure to air pollution, including CO, is responsible for millions of premature deaths each year, especially in urban areas where traffic and industrial emissions are high. These alarming figures highlight the need for effective monitoring and forecasting of air quality. Accurate prediction of CO levels can help governments, health organizations, and communities take preventive actions to reduce exposure and associated health risks.

Traditional air pollution monitoring relies on specialized sensors and equipment that continuously record pollutant levels. While effective, these systems are expensive to maintain and often limited to major urban centers. This leaves many smaller cities and rural areas without adequate monitoring infrastructure. In such cases, data-driven forecasting methods become highly valuable, as they can estimate pollutant trends using historical data without requiring costly real-time measurements everywhere. By applying statistical and machine learning models, researchers can analyze past CO concentration patterns to predict future values, which supports timely decision-making in environmental and health policy.

Over the past few years, several time series forecasting techniques have been applied to air quality prediction. Classical approaches such as the Auto-Regressive Integrated Moving Average (ARIMA) and Holt-Winters exponential smoothing have been widely used to capture temporal dependencies and seasonal variations [1]. More recently, machine learning methods, including Random Forests and deep learning architectures like Long Short-Term Memory (LSTM) networks, have shown promising results in handling complex, nonlinear relationships in environmental data [2], [3]. These models can take advantage of additional features such as lagged pollutant levels, rolling averages, and time-related variables (e.g., day of week, month) to improve prediction accuracy.

The main motivation for this study is to explore different forecasting approaches for predicting daily CO concentrations using a real-world dataset. Accurate CO forecasts are particularly important in regions where monitoring stations are sparse, as they can provide early warnings of pollution peaks and support public health interventions. The analysis in this study involved a sequence of steps: data preprocessing to ensure data quality, exploratory data analysis (EDA) to understand underlying patterns, feature engineering to enrich predictive information, model selection and training with three different forecasting techniques, and finally, evaluation and comparison of the models to determine the most effective forecasting method. To enable a comprehensive comparison, the study applied two statistical techniques such as ARIMA and Holt-Winters exponential smoothing, and one machine learning technique which is the Random Forest Regressor. By evaluating their performance using standard error metrics, we aimed to identify the most effective method for CO forecasting.

II. Methodology

• Dataset Description

The dataset used in this study consists of time series measurements of carbon monoxide (CO) concentrations, recorded at multiple intervals per day over a period of several years. The data is provided in CSV format and contains 8760 records with two primary attributes:

- **Timestamp:** Records the exact date and time at which the CO concentration was measured. This attribute provides the temporal dimension necessary for time series analysis. There are a total of 8760 values recorded here.
- **CO Level (ppm):** Represents the concentration of carbon monoxide in parts per million (ppm) at the corresponding timestamp. This variable serves as the target for forecasting models. This feature contains 6770 values and the remaining 1990 records don't have any value.

The dataset offers both fine-grained temporal resolution and long-term coverage, enabling the analysis of short-term fluctuations, seasonal patterns, and long-term trends in CO

concentrations. Such characteristics make it well-suited for evaluating different forecasting techniques and understanding their ability to capture the inherent variability in air pollution data.

● Data Preprocessing

Data preprocessing was carried out in several stages to ensure the quality, consistency, and suitability of the dataset for time series forecasting. The main steps included handling missing values, detecting and treating outliers, converting timestamps into a standard time series format, aggregating the data to daily averages, and performing initial visualization to assess temporal trends.

Handling Missing Values:

Missing observations in the dataset were addressed using a combination of forward filling and backward filling techniques. Specifically, missing values in the CO Level column were imputed using the most recent valid observation (`ffill()` function in python) and, where necessary, replaced by the next available observation (`bfill()` function in python). This approach ensured continuity in the time series without introducing artificial fluctuations, which is particularly important in air quality monitoring where gaps may distort temporal patterns.

Outlier Detection and Removal:

Outliers were identified using the interquartile range (IQR) method, a robust statistical technique commonly applied in environmental data analysis. The first quartile (Q1) and third quartile (Q3) of the CO Level distribution were calculated, and the interquartile range ($IQR = Q3 - Q1$) was used to define thresholds for plausible values. Observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were considered outliers and excluded from the cleaned dataset. This process reduced the influence of extreme values, which could otherwise bias the modeling results. A comparative plot of the original and cleaned datasets was generated to visualize the effect of outlier removal.

To evaluate the impact of this step, the time series was visualized in two forms: (i) the original series containing outliers, and (ii) the cleaned series with outliers removed. These comparative plots highlighted how extreme values were suppressed, leading to a smoother and more representative dataset.

Time Indexing and Aggregation:

To prepare the dataset for time series forecasting, the Timestamp column was converted into a DateTime index. The data was then resampled to a daily frequency, with the mean concentration of CO computed for each day. This step served two purposes: (i) it standardized the temporal resolution of the dataset, ensuring uniform intervals, and (ii) it reduced potential noise arising from high-frequency fluctuations, thereby highlighting long-term patterns more effectively.

Visualization:

A time series plot of the daily average CO concentrations was generated to provide an initial overview of the data. This visualization allowed for the inspection of underlying patterns, such

as potential seasonality, long-term trends, or irregular variations. Visual inspection also provided insight into the effectiveness of the preprocessing steps, particularly the removal of outliers and the handling of missing values.

Through this preprocessing pipeline, the dataset was transformed into a clean, consistent, and reliable time series, suitable for further analysis and forecasting.

● **Exploratory Data Analysis (EDA)**

To better understand the underlying characteristics of the CO concentration time series, exploratory data analysis was performed. This step involved decomposing the time series into its constituent components and conducting statistical tests to examine its stationarity, which is a key assumption for many forecasting models.

Time Series Decomposition (STL Method):

The time series was decomposed into three components: trend, seasonality, and residual, using the Seasonal-Trend decomposition procedure based on Loess (STL). STL is a widely used decomposition technique because it is highly flexible and can handle both additive and multiplicative seasonal effects. The decomposed components demonstrate below properties:

- **Trend component:** Represents the long-term progression in the data, capturing the overall direction of CO concentration levels over time.
- **Seasonal component:** Captures recurring patterns within a fixed period (e.g., daily, monthly, or yearly cycles), which are common in environmental datasets due to climatic and anthropogenic influences.
- **Residual component:** Represents the irregular fluctuations in the data after removing the trend and seasonal effects. These are often attributed to short-term variations or random noise.

By plotting the decomposed components, the long-term trend and periodic seasonal behavior of CO concentrations could be visualized, allowing for the identification of systematic patterns and irregular deviations.

Stationarity Testing (Augmented Dickey-Fuller Test):

To evaluate whether the CO concentration time series is stationary, the Augmented Dickey-Fuller (ADF) test was applied both to the raw series and to the residuals from the STL decomposition. Stationarity is a critical property in time series forecasting because many statistical models, such as ARIMA, assume that the mean and variance of the series remain constant over time.

The ADF test examines the null hypothesis that a unit root is present in the series (i.e., the series is non-stationary). If the test statistic is sufficiently negative and the p-value is below a

chosen significance threshold (commonly 0.05), the null hypothesis is rejected, indicating stationarity.

- **ADF test on residuals:** This was used to check whether the stochastic component of the decomposed series exhibited stationarity after removing systematic trend and seasonal effects.
- **ADF test on the original series:** This was used to determine whether the raw CO concentration data itself is stationary without decomposition.

The combined use of STL decomposition and the ADF test provided a comprehensive understanding of the temporal structure of the dataset, revealing both deterministic (trend and seasonality) and stochastic (residual) components. This analysis informed subsequent modeling decisions by clarifying whether differencing or detrending was required to achieve stationarity.

● Feature Engineering

Feature engineering was performed to extract meaningful information from the raw time series data and to enhance the predictive capacity of the forecasting models. Since time series data often exhibit temporal dependencies, seasonality, and long-term trends, carefully designed features are crucial to improving model performance. The following steps were implemented:

Time-based Features:

To capture temporal patterns, several time-related features were derived from the timestamp index of the dataset. These included the **day of the week**, **month**, **day of the month**, and **year**.

- **Day of the week** (0 = Monday, ..., 6 = Sunday): This feature helps identify weekly seasonality, such as differences between weekdays and weekends in pollutant levels.
- **Month:** Air pollutant concentrations are often influenced by seasonal variations (e.g., temperature and meteorological conditions). The month variable provides insight into seasonal cycles.
- **Day of the month and Year:** These features capture potential calendar effects and long-term trends across multiple years.
If the dataset contained hourly records, an additional **hour of the day** feature could be extracted to capture diurnal variations in CO levels, which are often influenced by traffic and industrial activities.

Lag Features:

Lag features were created to represent past observations of the CO time series, allowing models to capture autocorrelation in pollutant levels. Specifically:

- **Lag-1:** The CO concentration on the previous day. This captures short-term dependencies and persistence in pollutant concentrations.
- **Lag-7:** The CO concentration one week prior. This feature reflects weekly cyclic patterns, such as recurring traffic or industrial activities.
Including lagged variables provides the models with direct access to past information, which is a cornerstone of time series forecasting.

Rolling Statistics:

Rolling window statistics were calculated to incorporate smoothed information about local trends and variability.

- **7-day Rolling Mean and Standard Deviation:** These features capture short-term fluctuations and weekly trends while smoothing out noise. The rolling mean represents the local average CO level over the past week, while the rolling standard deviation measures short-term variability.
- **30-day Rolling Mean and Standard Deviation:** These features capture longer-term seasonal trends and variability across a monthly horizon. This helps account for broader seasonal effects, such as changes in weather conditions, heating activities, or industrial emissions.

By combining rolling means and rolling standard deviations, the models are provided with information not only about the central tendency of the pollutant levels but also their dispersion over different temporal windows.

Data Cleaning:

Since lag and rolling features inherently introduce missing values at the beginning of the series, these initial observations were removed to ensure a consistent dataset for modeling.

Feature Visualization:

To better understand the engineered features and their relationship with CO concentrations, visualizations were conducted.

- A **line plot** of CO levels alongside lagged values and rolling means was used to illustrate short- and long-term trends. This visualization highlights the smoothing effects of rolling statistics and the correlation between current and past pollutant values.

- A **boxplot of CO levels by day of the week** was used to detect weekly seasonality and variations in pollutant distribution across weekdays and weekends. Such visualization aids in identifying recurring weekly cycles that may influence air quality

• Model Selection and Training

In this study, three forecasting approaches were implemented to predict carbon monoxide (CO) concentration levels: two traditional statistical time series models: ARIMA and Holt-Winters exponential smoothing and one machine learning-based approach: Random Forest Regressor. Each model was trained on historical CO data and evaluated in terms of forecast accuracy using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

STL Decomposition and Residual Stationarity for Fitting Models:

Prior to modeling, the Timestamp data was decomposed into trend, seasonal, and residual components using **Seasonal-Trend decomposition via Loess (STL)**:

$$y_t = T_t + S_t + R_t \quad (1)$$

Where,

- y_t = Observed CO concentration at time t
- T_t = Trend component at time t
- S_t = Seasonal component at time t
- R_t = Residual (remainder) component at time t

Decomposing the series enabled the isolation of stationary residuals, which are more suitable for time series modeling. The residuals were then differenced to further ensure stationarity:

$$R'_t = R_t - R_{t-1} \quad (2)$$

Where,

- R'_t = Differenced residual at time t
- R_t = Residual at time t
- R_{t-1} = Residual at previous time step $t - 1$

Train-Test Split:

The processed data were split into training and testing sets using an 80:20 ratio. This split facilitated out-of-sample evaluation of the models and ensured that the test set simulated future predictions.

ARIMA Model:

The **Autoregressive Integrated Moving Average (ARIMA)** model is a parametric time series approach defined by three parameters: p , d , and q , representing the autoregressive order, differencing order, and moving average order, respectively. To determine appropriate values for p and q , the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the stationary residuals were examined through correlogram plots. Significant spikes in these plots indicate the presence of autocorrelations at specific lags, guiding the selection of the autoregressive (p) and moving average (q) terms. The differencing order (d) was determined based on stationarity tests, such as the Augmented Dickey-Fuller test, which indicates that the original series may or may not require differencing to achieve mean stationarity. By combining the insights from the ACF/PACF analysis and stationarity assessment, the ARIMA model was parametrized to effectively capture both linear dependencies and temporal structures in the residual component of the CO time series. The general ARIMA(p,d,q) model is expressed as:

$$\Phi_p(L)(1 - L)^d y_t = \Theta_q(L)\varepsilon_t \quad (3)$$

Where,

- $\Phi_p(L)$ = Autoregressive (AR) polynomial of order p
- L = Lag operator ($Ly_t = y_{t-1}$)
- d = Order of differencing required to make the series stationary
- $(1 - L)^d y_t$ = Differenced time series of order d
- $\Theta_q(L)$ = Moving Average (MA) polynomial of order q
- ε_t = White noise (error term) at time t
- y_t = Observed value at time t

In this study, the ARIMA model was applied directly to the differenced residuals obtained from STL decomposition, effectively modeling the stationary component of the series.

Holt-Winters Exponential Smoothing:

Holt-Winters exponential smoothing is a statistical forecasting method that captures level, trend, and seasonal components. The additive Holt-Winters equations are defined as:

$$L_t = \alpha(y_t - S_{t-m}) + (1 - \alpha)(L_{t-1} + T_{t-1}) \quad (4)$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \quad (5)$$

$$S_t = \gamma(y_t - L_t) + (1 - \gamma)S_{t-m} \quad (6)$$

$$\hat{y}_{t+h} = L_t + hT_t + S_{t-m+h_m} \quad (7)$$

Where,

- L_t = Level component at time t
- T_t = Trend component at time t
- S_t = Seasonal component at time t
- y_t = Observed value at time t
- α = Smoothing parameter for level ($0 \leq \alpha \leq 1$)
- β = Smoothing parameter for trend ($0 \leq \beta \leq 1$)
- γ = Smoothing parameter for seasonality ($0 \leq \gamma \leq 1$)
- m = Seasonal period (e.g., 7 for weekly seasonality in daily data)
- $\hat{y}_{t+h}$ = Forecasted value h steps ahead
- h_m = Adjustment to align seasonal component with forecast horizon

In this analysis, Holt-Winters was fitted to the residuals to capture any remaining systematic structure after trend and seasonal decomposition.

Random Forest Regressor:

This is a non-parametric ensemble machine learning method based on decision trees. For time series forecasting, lagged observations and rolling statistics were used as input features to capture temporal dependencies. Specifically, features included:

- Lag values (1-day lag, 7-days lag)
- Rolling mean and standard deviation over 7-day and 30-day windows

The model estimates the target residual as:

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^B f_b(X_t) \quad (8)$$

Where,

- $\hat{y}_t$ = Predicted value of residual at time t
- B = Total number of decision trees in the forest
- $f_b(X_t)$ = Prediction of the b -th tree given input features X_t
- X_t = Feature vector at time t (e.g., lag values, rolling mean, rolling std)

Random Forest is robust to non-linearities and can automatically capture complex interactions among features without explicit parametric assumptions.

Forecast Reconstruction:

Forecasts from each model were combined with the trend and seasonal components obtained from STL decomposition to reconstruct CO predictions on the original scale:

$$\hat{y}_t^{\text{forecast}} = \hat{R}_t + T_t + S_t \quad (9)$$

Where,

- $\hat{y}_t^{\text{forecast}}$ = Final forecasted CO level at time t
- $\hat{R}_t$ = Forecasted residual at time t from the model
- T_t = Trend component at time t
- S_t = Seasonal component at time t

This step ensured that the final forecasts reflected both long-term patterns and stochastic fluctuations observed in the data.

Model Evaluation:

Model performance was evaluated using three metrics: **mean absolute error (MAE)**, **root mean squared error (RMSE)**, and **mean absolute percentage error (MAPE)**:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (11)$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (12)$$

Where,

- n = Number of observations
- y_t = Actual value at time t
- $\hat{y}_t$ = Predicted value at time t

Missing values were handled to ensure accurate calculation. Visualizations of both reconstructed forecasts and error metrics facilitated a comparative assessment of model performance.

• Forecasting and Visualization

Following model evaluation, the best-performing forecasting model was selected to predict future CO concentration levels. The selected model was used to generate forecasts for the subsequent time period (e.g., the next month), extending beyond the historical data to provide actionable predictions for environmental monitoring.

The forecasted values were plotted alongside historical observations to enable a visual assessment of the model's predictive accuracy. This visualization defines how well the model captured both seasonal patterns and residual variations while providing insight into potential future trends. Plotting the forecasts in the context of historical data also facilitated the identification of periods where predictions might be less reliable due to unusual fluctuations or anomalies in the time series.

In addition to graphical representation, a tabular summary of the best performing model's forecasted CO concentrations was produced. This table provided numerical estimates for each future time point, allowing stakeholders and policymakers to directly interpret predicted pollutant levels and plan mitigation or alert strategies accordingly.

By combining visual and tabular outputs, this section ensured that both qualitative and quantitative assessments of the forecast were available, supporting informed decision-making for public health and environmental management.

III. Result and Discussion

Dataset Visualization and EDA:

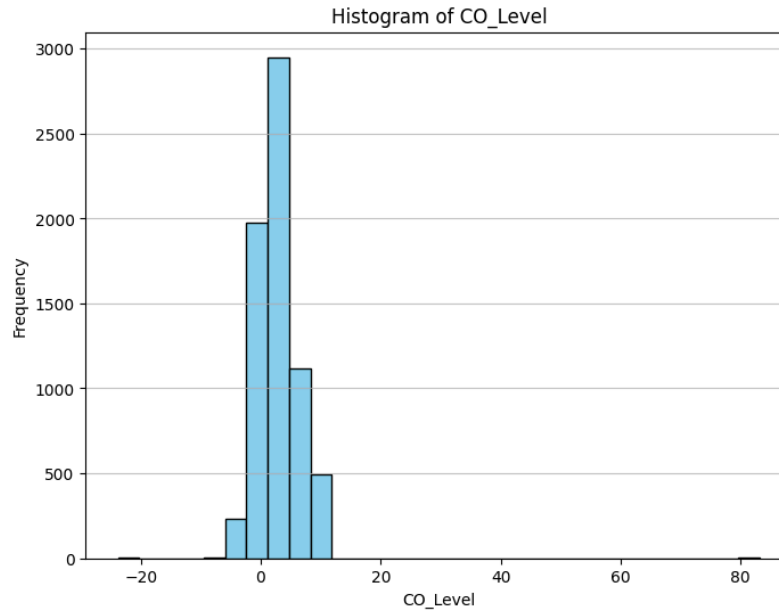


Figure 1: Data distribution plot of CO Level

Figure 1 indicates that the bulk of daily CO measurements are approximately normally distributed, with two conspicuous extreme values (one unusually low, one unusually high). From a modeling perspective, this distributional shape supports the use of additive models (errors that do not scale multiplicatively with the level). However, the presence of isolated extremes motivates robust treatment because extreme sensor readings can disproportionately bias parameter estimates and error metrics.

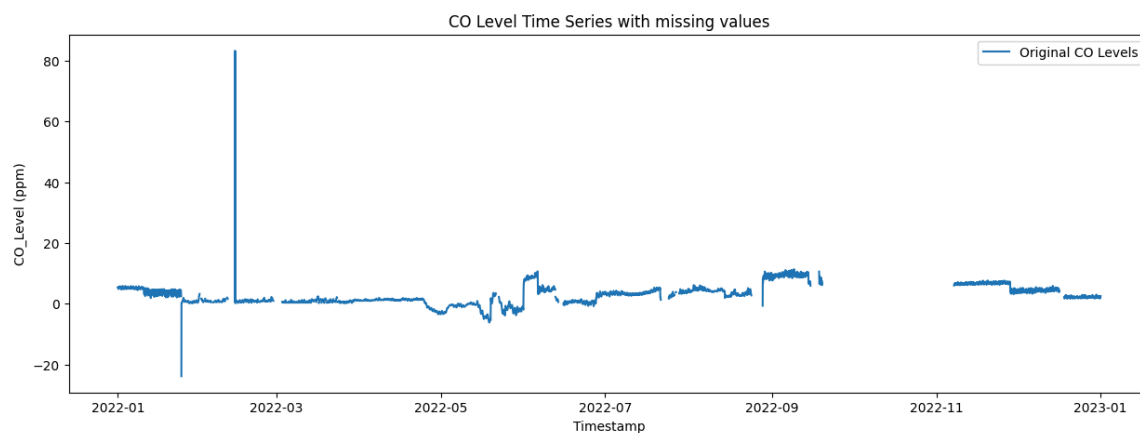


Figure 2: CO Levels Over Time

The raw hourly series in Figure 2 reveals several important data-quality issues: (1) intermittent missing values (visible as gaps), and (2) abrupt spikes that are likely outliers or event-driven extremes. Missingness at irregular times can distort seasonal estimates and autocorrelations unless handled carefully. Spikes may represent sensor faults which should be corrected. Hence, any outlier-removal strategy must balance removing measurement error and preserving genuine extreme events that are meaningful for public-health forecasting.

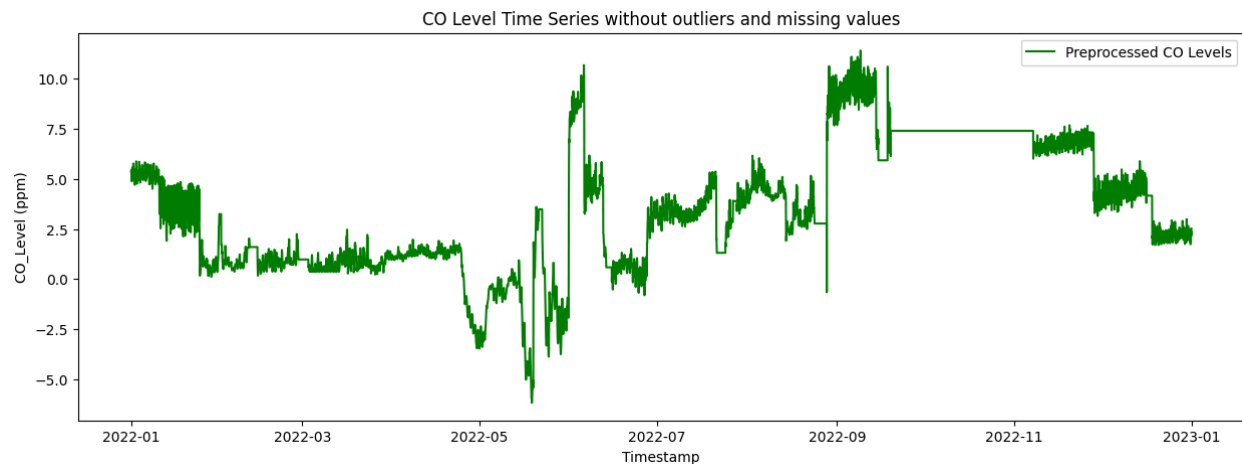


Figure 3: Preprocessed CO Levels Over Time

Figure 3 shows the dataset after preprocessing, where missing values were imputed and outliers were treated using the interquartile range (IQR) method. Straight segments in the plot reflect imputed values replacing missing observations, while the absence of extreme spikes indicates that anomalous values have been removed or corrected. The cleaned series also highlights the temporal dynamics of CO concentrations: rather than a simple monotonic trend, the data exhibit multiple structural shifts, such as abrupt increases around mid-2022 and early September, interspersed with relatively stable periods. No strong evidence of repeating seasonal cycles is apparent, though shorter fluctuations suggest the possibility of short-term periodic behavior. Importantly, the mean and variance are not constant over time, implying that the series is non-stationary. Given the relatively stable fluctuation size, the data are more consistent with an additive structure than a multiplicative one. These insights motivate the use of transformations such as differencing or formal decomposition methods (e.g., STL) in later analysis.

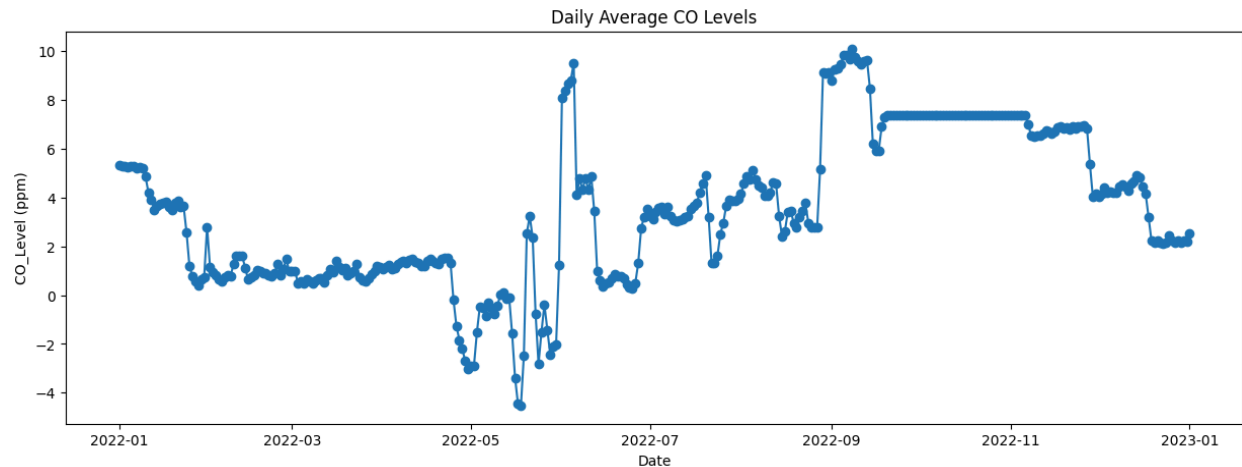


Figure 4: Average CO Levels Over Time (Day).

To reduce high-frequency noise, the data were resampled to daily averages, as illustrated in Figure 4. Aggregating the data to daily frequency produced a smoother series, emphasizing long-term movements and reducing the volatility observed in hourly measurements. This representation better reflects daily exposure levels and is more suitable for modeling broader temporal patterns.

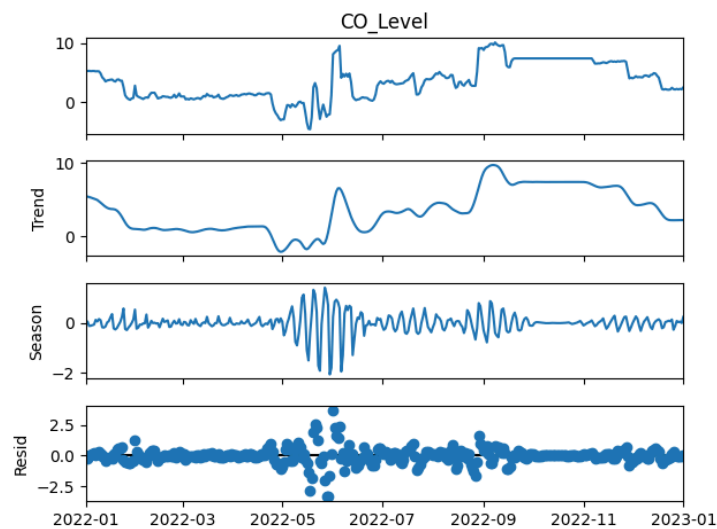


Figure 5: STL Decomposition of CO Level

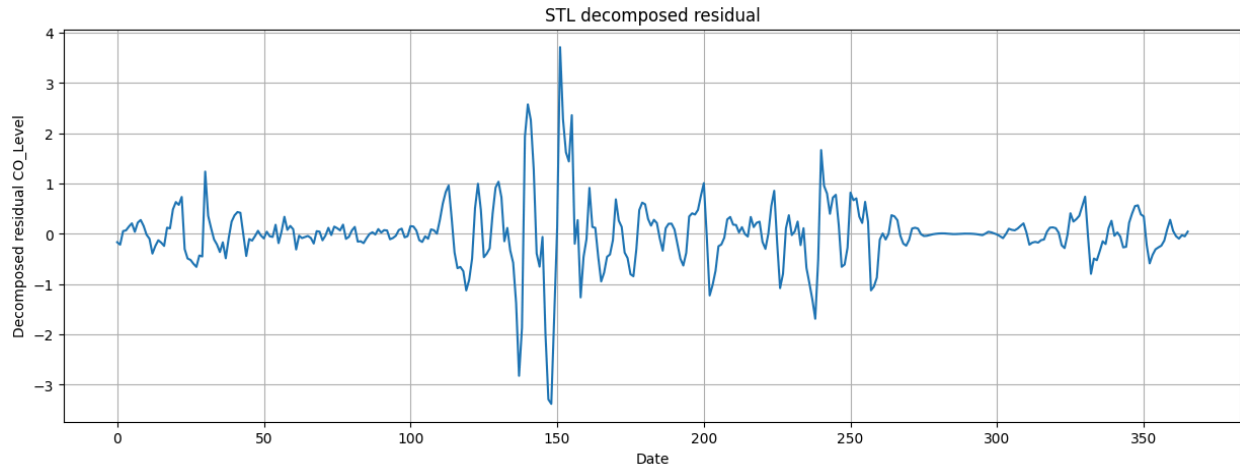


Figure 6: Residual component of STL decomposition

Figure 5 displays the decomposition of the daily CO concentration series into trend, seasonal, and residual components using Seasonal-Trend decomposition via Loess (STL). The trend component reflects structural shifts in the data rather than a continuous upward or downward trajectory, consistent with changes in environmental conditions or anthropogenic activity. The seasonal component suggests the presence of short-lived periodic cycles, but without a clearly persistent repeating pattern across the full time horizon. The residual component captures short-term fluctuations unexplained by the trend or seasonality. The residuals are visualized in both line and scatter plots in Figures 5 and 6 respectively. They fluctuate around zero, as expected, confirming that the trend and seasonal effects have been effectively removed. However, closer inspection reveals irregular bursts of volatility, particularly around mid-2022, where clusters of unusually high variability appear. This behavior indicates heteroskedasticity, meaning that the variance of the process is not constant over time. While the residuals are stationary in mean (centered around zero, with no evident trend), they are weakly non-stationary in variance due to volatility clustering. These findings suggest the presence of random shocks and highlight the potential need for models capable of addressing time-varying variance.

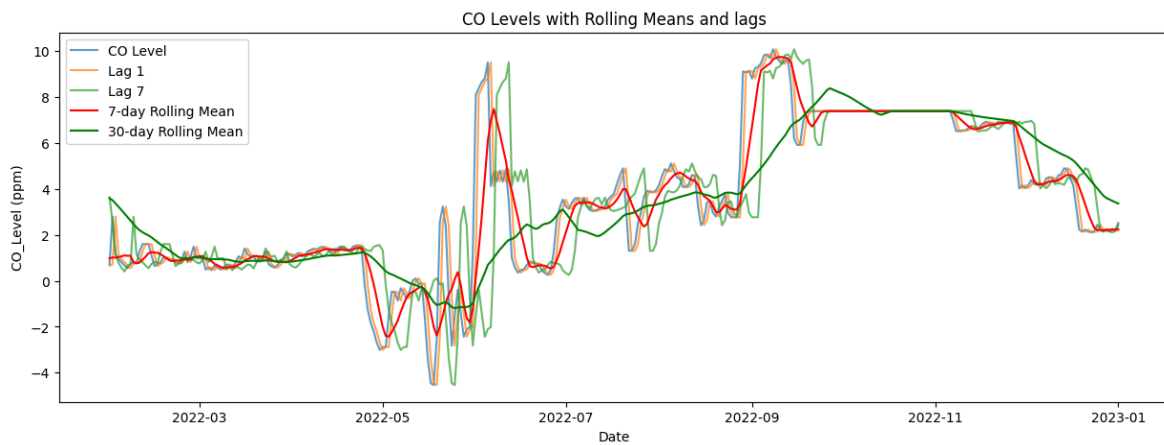


Figure 7: Lag and Rolling Statistics Features of CO Concentrations

Figure 7 visualizes derived features from the CO concentration time series to enhance forecasting of Random Forest Regressor. Lag features (Lag-1 and Lag-7) capture short-term persistence and weekly cycles in pollutant levels. Rolling window statistics (7-day and 30-day means and standard deviations) summarize local and long-term trends while measuring variability. Together, these features provide a richer temporal context by incorporating both past observations and smoothed patterns of change in CO concentrations.

ADF Test Result:

The **Augmented Dickey-Fuller (ADF) test** was conducted to assess the stationarity of the CO concentration time series, both for the preprocessed original data and for the residual component obtained from STL decomposition. Stationarity is a critical assumption for many time series models, including ARIMA, as it ensures that the statistical properties (mean, variance, autocorrelation) of the series remain constant over time.

Result of preprocessed CO Level:

ADF Statistic: -2.005344221740884

p-value: 0.2842347716438276

Result of decomposed STL residual:

ADF Statistic: -6.489684351497461

p-value: 1.23554829472552e-08

Explanation:

For the preprocessed CO levels, the ADF test produced a statistic of -2.005 with a corresponding p-value of 0.2840. Since the p-value exceeds the common significance threshold of 0.05, the null hypothesis of a unit root could not be rejected. This indicates that the original preprocessed series is non-stationary, with mean and/or variance that change over time. The non-stationarity is consistent with the earlier observations from exploratory data analysis, which revealed structural shifts and changing variance in the time series.

In contrast, the **residual component from STL decomposition** exhibited a substantially more negative ADF statistic of -6.49 and a p-value of 1.24×10^{-8} . This p-value is far below 0.05, leading to rejection of the null hypothesis of a unit root. Consequently, the residual series can be considered **stationary in the mean**, meaning it fluctuates around a constant mean over time.

While heteroskedasticity remains in the residuals (since variance is not strictly constant), the stationarity in mean confirms that the stochastic component is suitable for modeling with time series techniques such as ARIMA or machine learning models that assume stationarity.

Overall, these results justify the decomposition approach which removes trend and seasonal components and transforms the original non-stationary series into a residual component that satisfies the stationarity requirement for many forecasting models, improving both interpretability and predictive accuracy.

ACF and PACF Correlogram plots of the STL decomposed residual:

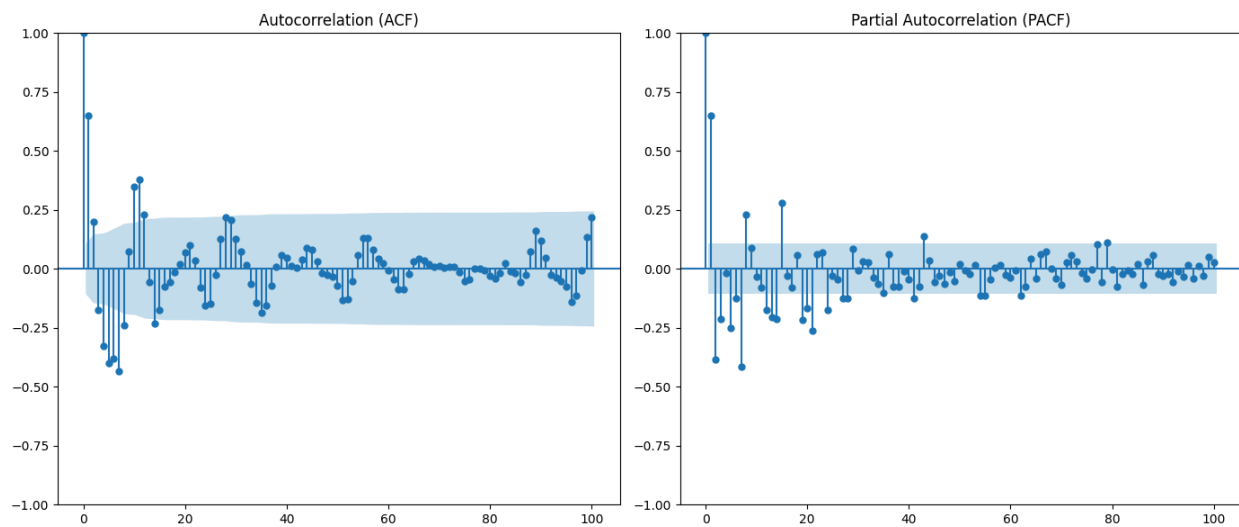


Figure 8: Correlogram plots of ACF and PACF

To select appropriate parameters for the ARIMA model, the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the STL-decomposed residuals were first examined using correlogram plots of Figure 8. The ACF plot measures the correlation between the time series and its lagged values, while the PACF isolates the direct effect of each lag by removing the influence of intervening lags. Significant spikes in these plots indicate lags at which autocorrelations are statistically meaningful, providing guidance for the autoregressive (p) and moving average (q) components of the ARIMA model.

Analysis of the correlogram plots revealed that both the ACF and PACF exhibited significant spikes up to lag 2. In conjunction with the Augmented Dickey-Fuller test results, which indicated that the STL residuals were stationary ($d = 0$), these observations were used to determine the ARIMA model parameters as $p = 2$, $d = 0$, and $q = 2$. This configuration allows the model to effectively capture the linear dependencies and short-term autocorrelations present in the stationary residuals, ensuring accurate modeling of the stochastic component of CO concentrations.

Model Performance Evaluation and Comparison:

The forecasting performance of the three models (ARIMA, Holt-Winters exponential smoothing, and Random Forest Regressor) was assessed using three commonly used error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics quantify the deviation between predicted and actual CO concentrations, with lower values indicating better predictive accuracy.

Table 1: Result of the Performance Error Metrics of ARIMA, Holt-Winters, and Random Forest Models

Models	MAE	RMSE	MAPE
ARIMA	0.209871	0.279181	5.055122
Holt-Winters Exponential Smoothing	0.215040	0.282679	5.282048
Random Forest Regressor	0.180741	0.244424	4.643877

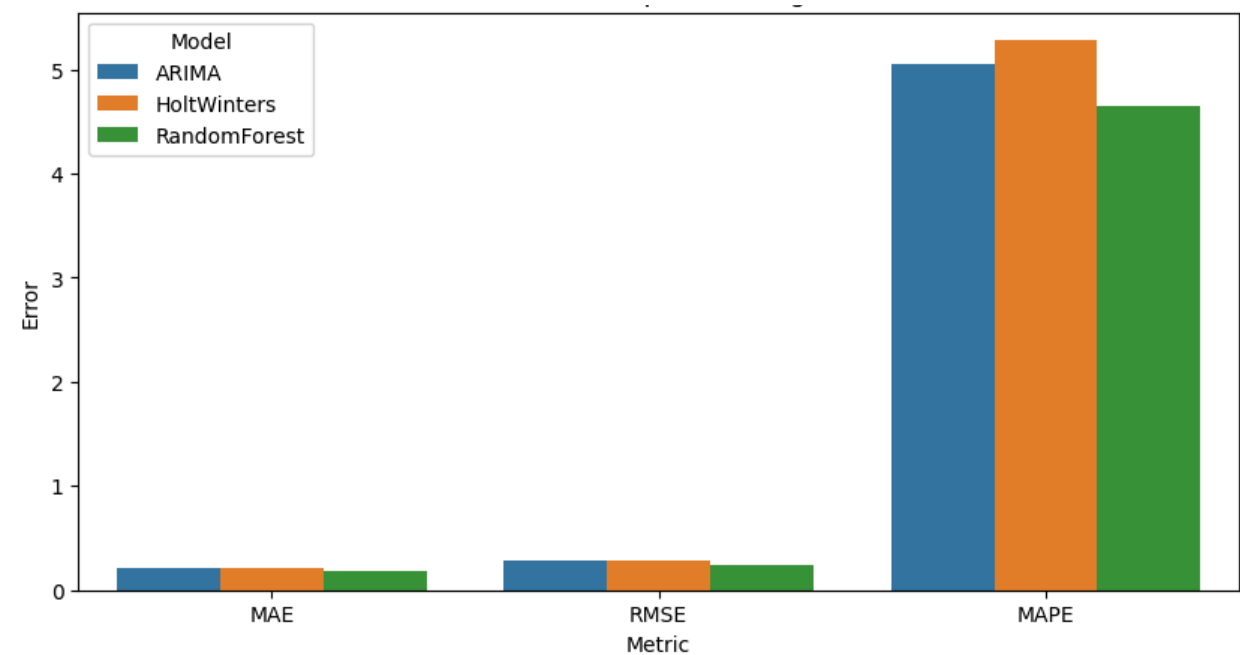


Figure 9: Models’ Result of the Performance Error Metrics.

The evaluation results are recorded in Table 1 and compared in Figure 9. The **Random Forest Regressor** outperformed both statistical models across all three metrics, achieving an MAE of 0.181, an RMSE of 0.244, and a MAPE of 4.64%. This indicates that the Random Forest was more effective at capturing both short-term fluctuations and complex non-linear relationships in the residual component of the CO time series.

The **ARIMA model** demonstrated slightly better performance than Holt-Winters, with an MAE of 0.210, RMSE of 0.279, and MAPE of 5.06%. This suggests that ARIMA is capable of modeling the stationary residuals effectively, capturing autocorrelations and linear dependencies in the data.

The **Holt-Winters exponential smoothing** method exhibited the highest errors among the three models (MAE = 0.215, RMSE = 0.283, MAPE = 5.28%). While Holt-Winters is efficient in capturing trend and seasonality in time series, its performance is slightly limited in this context, likely because the residuals exhibit irregular fluctuations and short-term volatility that are less well modeled by simple exponential smoothing.

Overall, these results indicate that machine learning-based approaches, such as Random Forest, can provide improved predictive accuracy for complex time series residuals, particularly when the data exhibit non-linearities, heteroskedasticity, or irregular short-term patterns. The statistical models ARIMA and Holt-Winters still offer reasonable performance, especially in capturing linear correlations and structured seasonal patterns, but may be less flexible in handling stochastic variability.

Forecasting and Visualization:

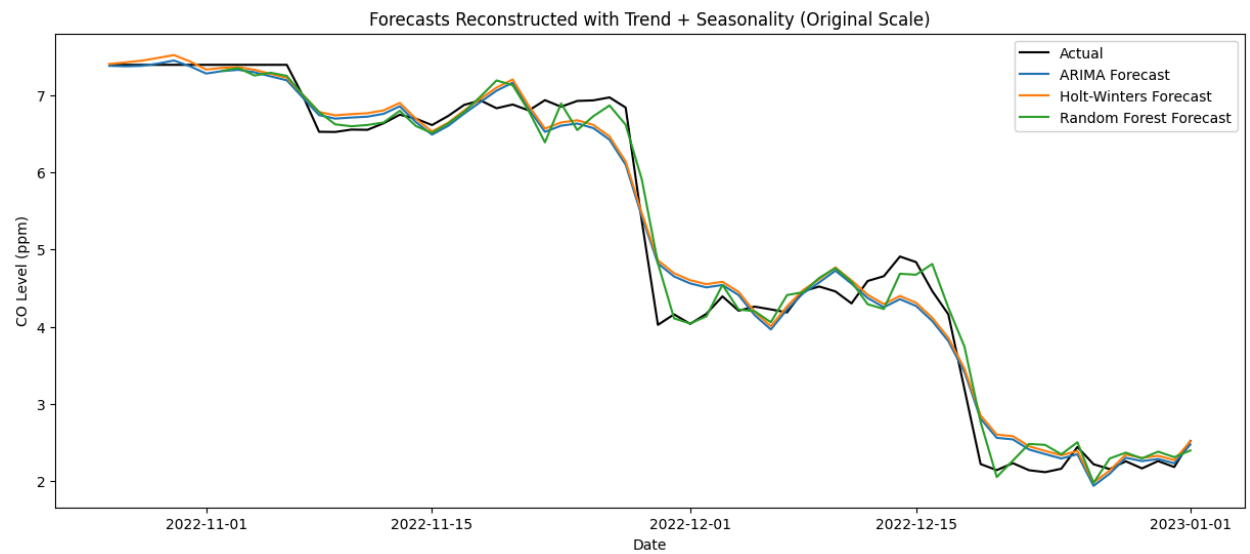


Figure 10: Models’ Forecasted vs. Actual CO Levels on Test Set.

Figure 10 presents a comparative visualization of the predicted CO concentrations from the three models—ARIMA, Holt-Winters exponential smoothing, and Random Forest Regressor—against the actual observed values in the test set. The plot illustrates how closely each model tracks the true CO levels. Consistent with the numerical evaluation, the Random Forest predictions align most closely with the observed values, capturing both short-term fluctuations and abrupt changes in the series. ARIMA shows reasonable accuracy but occasionally underestimates sharp deviations, while Holt-Winters exhibits slightly higher

deviations, particularly during periods of irregular spikes in CO concentration. Overall, the figure highlights the superior predictive capability of the Random Forest model in modeling complex residual patterns.

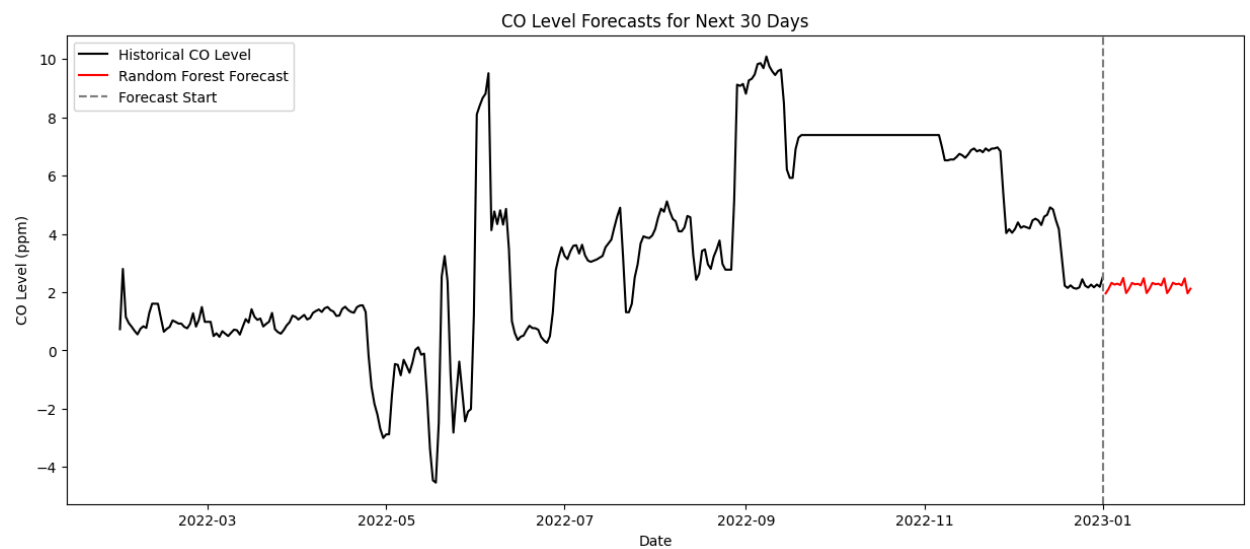


Figure 11: Random Forest Regressor’s Forecasted CO Levels for next 30 days.

Figure 11 depicts the Random Forest Regressor’s forecast for CO concentrations over the subsequent 30-day period. The plot demonstrates the model’s ability to project future trends based on learned patterns in the historical data. The forecasted series shows expected short-term variability while maintaining consistency with the overall level of the test data. This visualization provides a practical assessment of potential future CO exposure and can serve as an early warning tool for environmental monitoring and public health planning.

Table 2: Random Forest Regressor’s Forecasted CO Levels for Next 30 Days

Date	CO Level
2023-01-02	1.956
2023-01-03	2.112
2023-01-04	2.319
2023-01-05	2.258
2023-01-06	2.290
2023-01-07	2.242
2023-01-08	2.486

2023-01-09	1.967
2023-01-10	2.117
2023-01-11	2.317
2023-01-12	2.266
2023-01-13	2.285
2023-01-14	2.233
2023-01-15	2.476
2023-01-16	1.967
2023-01-17	2.119
2023-01-18	2.318
2023-01-19	2.268
2023-01-20	2.289
2023-01-21	2.229
2023-01-22	2.471
2023-01-23	1.965
2023-01-24	2.120
2023-01-25	2.323
2023-01-26	2.269
2023-01-27	2.289
2023-01-28	2.229
2023-01-29	2.471

Table 2 provides a detailed numerical summary of the Random Forest Regressor's forecasted CO concentrations for each day over the 30-day prediction horizon. These values can be directly utilized for planning mitigation strategies or issuing alerts. The table complements Figure 11 by offering exact forecasted levels, allowing quantitative interpretation and further analysis, such as comparison with regulatory thresholds or historical averages.

Discussion:

During the study, several challenges were encountered. First, the dataset contained a large number of missing values and outliers, which required careful preprocessing to avoid biasing the models. Handling non-stationarity was also a key issue, as the raw series exhibited structural shifts and changing variance. Although STL decomposition and differencing addressed mean stationarity, heteroskedasticity remained a challenge, particularly for statistical models like Holt-Winters.

Potential improvements could include experimenting with more advanced machine learning and deep learning models such as Gradient Boosting Machines (XGBoost, LightGBM) or Long Short-Term Memory (LSTM) networks, which are better equipped to capture complex temporal dependencies and volatility. Incorporating exogenous variables, such as meteorological data (temperature, wind speed, humidity), could also enhance predictive accuracy by accounting for external factors that influence air pollution levels.

The implications of this study extend to real-world scenarios such as climate studies, environmental monitoring, and public health planning. Accurate forecasts of CO concentrations can support early warning systems, inform traffic or industrial regulations, and help policymakers design interventions to reduce exposure risks. More broadly, the methodology demonstrated, combining statistical and machine learning approaches can be applied to forecasting other pollutants or climate indicators, enabling data-driven responses to environmental challenges.

IV. Conclusion

This study investigated the forecasting of daily carbon monoxide concentrations using ARIMA, Holt-Winters exponential smoothing, and Random Forest Regressor models. The preprocessing pipeline successfully transformed a noisy and incomplete dataset into a usable time series through imputation, outlier removal, and decomposition. Comparative evaluation showed that the Random Forest Regressor achieved superior performance across all error metrics, demonstrating its ability to capture nonlinear and short-term fluctuations more effectively than traditional statistical models. Overall, the work highlights the effectiveness of integrating classical time series techniques with machine learning to improve forecasting accuracy in environmental studies. The findings not only provide insights into CO pollution dynamics but also underscore the broader applicability of such hybrid approaches for air quality management, climate research, and environmental policy support.

V. References

[1] A. D. Syafei, N. Ramadhan, J. Hermana, A. Slamet, R. Boedisantoso, and A. F. Assomadi, "Application of Exponential Smoothing Holt Winter and ARIMA Models for Predicting Air Pollutant Concentrations," *EnvironmentAsia*, vol. 11, no. 3, pp. 251–262, 2018, doi: 10.14456/ea.2018.52. tshe.org

[2] Y. Wang, P. Liu, C. Xu, C. Peng, and J. Wu, "A deep learning approach to real-time CO concentration prediction at signalized intersection," *Atmospheric Pollution Research*, vol. 11, no. 8, pp. 1370–1378, 2020, doi: 10.1016/j.apr.2020.05.007. [OUOI](#)

[3] R. Navares and J. L. Aznarte, "Predicting air quality with deep learning LSTM: Towards comprehensive models," *Ecological Informatics*, vol. 55, Art. no. 101019, 2020, doi: 10.1016/j.ecoinf.2019.101019.